

Hypothesis 4 - Emotional Condition × Cognitive Load on Working-Memory Performance

Confirmatory Hypothesis Report

Hypothesis

The effect of emotional condition on working-memory performance varies according to cognitive load across the 1-back, 2-back, 3-back, and 4-back levels.

Method

A confirmatory interaction analysis was conducted to evaluate whether the effect of emotional condition on working-memory performance varied as a function of cognitive load. The analytical sample included 23 participants.

Emotional condition comprised Neutral, Positive, and Negative levels, while cognitive load was represented by the 1-back, 2-back, 3-back, and 4-back task levels. The primary inferential term was the interaction between emotional condition and n-back level.

Response accuracy was analyzed using a generalized estimating equation (GEE) model with a binomial distribution, logit link, exchangeable within-participant correlation structure, and robust standard errors. The accuracy analysis included 17,820 trial-level observations.

Reaction time was analyzed among valid correct-response trials. Because reaction-time values were positively skewed, reaction time was log-transformed. A linear mixed-effects model was fitted with emotional condition, n-back level, and their interaction as fixed effects, together with participant-specific random intercepts. The reaction-time analysis included 14,184 observations.

The global Emotion × N-back interaction for accuracy was evaluated using a Wald test. For reaction time, the full interaction mixed-effects model was compared with an additive model using a likelihood-ratio test under maximum likelihood estimation. Significant interactions were followed by planned simple-effect analyses comparing Positive versus Neutral, Negative versus Neutral, and Positive versus Negative at each n-back level.

GEE coefficients were exponentiated and interpreted as odds ratios. Log reaction-time coefficients were converted to percentage changes in reaction time. Statistical significance was evaluated using a two-sided alpha level of .05, with 95% confidence intervals reported.

Analytical Sample

The distribution of participants and trials across emotional conditions and n-back levels is summarized below.

Table 1. Analytical sample by emotional condition and cognitive-load level.

Condition	N-back	Participants	Trials	Valid accuracy
Negative	1	22	1320	1320
Negative	2	22	1980	1980
Negative	3	22	1320	1320
Negative	4	22	1320	1320
Neutral	1	23	1320	1320
Neutral	2	23	1980	1980

Condition	N-back	Participants	Trials	Valid accuracy
Neutral	3	23	1320	1320
Neutral	4	23	1320	1320
Positive	1	22	1320	1320
Positive	2	22	1980	1980
Positive	3	22	1320	1320
Positive	4	22	1320	1320

Results

Accuracy

The global Emotion $\times$ N-back interaction for response accuracy was statistically significant ($p < .001$). This indicates that the association between emotional condition and accuracy varied across levels of cognitive load.

Among the interaction coefficients, 2 terms reached the predefined alpha level of .05.

The descriptive and simple-effect analyses were therefore used to characterize how the emotional-condition contrasts changed from 1-back through 4-back. Because interaction coefficients are conditional on the model reference levels, interpretation focuses primarily on the planned simple effects within each cognitive-load level.

At the 1-back level, the following planned contrast(s) were statistically significant: Negative vs Neutral (OR = 0.60, $p = .025$); Positive vs Negative (OR = 2.91, $p = .005$).

At the 4-back level, the following planned contrast(s) were statistically significant: Negative vs Neutral (OR = 0.76, $p = .007$).

Table 2. Descriptive accuracy by emotional condition and n-back level.

Condition	N-back	Participants	Trials	Correct	Accuracy	SD	Median
Negative	1	22	1320	1289	97.7%	0.151	1.0
Negative	2	22	1980	1764	89.1%	0.312	1.0
Negative	3	22	1320	1018	77.1%	0.420	1.0
Negative	4	22	1320	907	68.7%	0.464	1.0
Neutral	1	23	1320	1301	98.6%	0.119	1.0
Neutral	2	23	1980	1774	89.6%	0.305	1.0
Neutral	3	23	1320	990	75.0%	0.433	1.0
Neutral	4	23	1320	984	74.5%	0.436	1.0
Positive	1	22	1320	1309	99.2%	0.091	1.0
Positive	2	22	1980	1779	89.8%	0.302	1.0
Positive	3	22	1320	1019	77.2%	0.420	1.0
Positive	4	22	1320	948	71.8%	0.450	1.0

Table 3. Emotion $\times$ cognitive-load interaction coefficients for response accuracy.

Term	Beta	OR	95% CI	p
C(emotion_condition, Treatment(reference='Neutral'))[T.Negative]:C(nback_level, Treatment(reference=1))[T.2]	0.431	1.539	[1.00, 2.37]	= .050
C(emotion_condition, Treatment(reference='Neutral'))[T.Positive]:C(nback_level, Treatment(reference=1))[T.2]	-0.498	0.608	[0.31, 1.20]	= .150
C(emotion_condition, Treatment(reference='Neutral'))[T.Negative]:C(nback_level, Treatment(reference=1))[T.3]	0.598	1.819	[1.13, 2.92]	= .013
C(emotion_condition, Treatment(reference='Neutral'))[T.Positive]:C(nback_level, Treatment(reference=1))[T.3]	-0.403	0.668	[0.33, 1.37]	= .273
C(emotion_condition, Treatment(reference='Neutral'))[T.Negative]:C(nback_level, Treatment(reference=1))[T.4]	0.194	1.215	[0.72, 2.04]	= .464
C(emotion_condition, Treatment(reference='Neutral'))[T.Positive]:C(nback_level, Treatment(reference=1))[T.4]	-0.662	0.516	[0.25, 1.07]	= .074

Table 4. Planned simple effects for response accuracy at each n-back level.

N-back	Contrast	OR	95% CI	p
1	Negative vs Neutral	0.603	[0.39, 0.94]	= .025
1	Positive vs Negative	2.910	[1.38, 6.13]	= .005
1	Positive vs Neutral	1.756	[0.87, 3.55]	= .117
2	Negative vs Neutral	0.952	[0.77, 1.18]	= .653
2	Positive vs Negative	1.076	[0.86, 1.35]	= .530
2	Positive vs Neutral	1.024	[0.89, 1.18]	= .743
3	Negative vs Neutral	1.128	[0.92, 1.38]	= .241
3	Positive vs Negative	0.995	[0.82, 1.21]	= .959
3	Positive vs Neutral	1.122	[0.94, 1.35]	= .211
4	Negative vs Neutral	0.758	[0.62, 0.93]	= .007
4	Positive vs Negative	1.139	[0.95, 1.36]	= .151
4	Positive vs Neutral	0.863	[0.70, 1.06]	= .167

Reaction Time

The global Emotion $\times$ N-back interaction for reaction time was not statistically significant (likelihood-ratio chi-square(6) = 11.78, $p = .067$).

A total of 2 individual reaction-time interaction terms reached the predefined alpha level.

Simple-effect analyses examined the percentage change in reaction time for each emotional-condition contrast at each n-back level. Negative percentage values indicate faster responding for the first condition in the contrast, whereas positive values indicate slower responding.

At the 1-back level, the following planned contrast(s) were statistically significant: Negative vs Neutral (RT change = -19.6%, $p < .001$); Positive vs Neutral (RT change = -22.8%, $p < .001$).

At the 2-back level, the following planned contrast(s) were statistically significant: Negative vs Neutral (RT change = -24.8%, $p < .001$); Positive vs Neutral (RT change = -26.9%, $p < .001$).

At the 3-back level, the following planned contrast(s) were statistically significant: Negative vs Neutral (RT change = -29.3%, $p < .001$); Positive vs Neutral (RT change = -29.3%, $p < .001$).

At the 4-back level, the following planned contrast(s) were statistically significant: Negative vs Neutral (RT change = -19.9%, $p < .001$); Positive vs Neutral (RT change = -21.6%, $p < .001$).

Table 5. Descriptive reaction time by emotional condition and n-back level.

Condition	N-back	Participants	Trials	Mean RT	SD	Median RT	Mean log RT
Negative	1	22	1191	487.14	403.11	347.00	5.97
Negative	2	22	1680	800.85	752.96	500.50	6.34
Negative	3	22	977	845.42	858.59	451.00	6.34
Negative	4	22	856	853.44	862.38	483.50	6.37
Neutral	1	23	1244	643.46	555.73	437.50	6.19
Neutral	2	23	1709	1036.54	900.37	720.00	6.61
Neutral	3	23	943	1172.93	1007.66	829.00	6.69
Neutral	4	23	933	1060.72	1018.84	661.00	6.58
Positive	1	22	1166	535.90	573.71	331.00	5.96
Positive	2	22	1653	821.82	817.70	475.00	6.32
Positive	3	22	961	869.15	909.86	477.00	6.34
Positive	4	22	871	882.73	878.01	532.00	6.37

Table 6. Emotion $\times$ cognitive-load interaction coefficients for log reaction time.

Term	Beta	RT change	95% CI	p
C(emotion_condition, Treatment(reference='Neutral'))[T.Negative]:C(nback_level, Treatment(reference=1))[T.2]	-0.055	-5.4%	[-12.3%, 2.0%]	= .149

Term	Beta	RT change	95% CI	p
C(emotion_condition, Treatment(reference='Neutral'))[T.Positive]:C(nback_level, Treatment(reference=1))[T.2]	-0.065	-6.3%	[-13.1%, 1.1%]	= .093
C(emotion_condition, Treatment(reference='Neutral'))[T.Negative]:C(nback_level, Treatment(reference=1))[T.3]	-0.112	-10.6%	[-18.0%, -2.5%]	= .011
C(emotion_condition, Treatment(reference='Neutral'))[T.Positive]:C(nback_level, Treatment(reference=1))[T.3]	-0.103	-9.7%	[-17.3%, -1.6%]	= .021
C(emotion_condition, Treatment(reference='Neutral'))[T.Negative]:C(nback_level, Treatment(reference=1))[T.4]	0.010	1.0%	[-7.5%, 10.3%]	= .822
C(emotion_condition, Treatment(reference='Neutral'))[T.Positive]:C(nback_level, Treatment(reference=1))[T.4]	0.006	0.6%	[-7.9%, 9.9%]	= .892

Table 7. Planned simple effects for reaction time at each n-back level.

N-back	Contrast	RT change	95% CI	p
1	Negative vs Neutral	-19.6%	[-23.5%, -15.5%]	< .001
1	Positive vs Negative	-4.0%	[-8.8%, 1.1%]	= .120
1	Positive vs Neutral	-22.8%	[-26.6%, -18.8%]	< .001
2	Negative vs Neutral	-24.8%	[-28.3%, -21.0%]	< .001
2	Positive vs Negative	-2.8%	[-7.6%, 2.1%]	= .254
2	Positive vs Neutral	-26.9%	[-30.4%, -23.2%]	< .001
3	Negative vs Neutral	-29.3%	[-33.9%, -24.4%]	< .001
3	Positive vs Negative	-0.0%	[-6.6%, 6.9%]	= .989
3	Positive vs Neutral	-29.3%	[-33.9%, -24.4%]	< .001
4	Negative vs Neutral	-19.9%	[-25.4%, -14.1%]	< .001
4	Positive vs Negative	-2.1%	[-8.9%, 5.3%]	= .572
4	Positive vs Neutral	-21.6%	[-26.9%, -15.9%]	< .001

Global Interaction Summary

The global interaction tests summarize whether emotional effects varied across cognitive-load levels for each primary outcome.

Table 8. Summary of global Emotion $\times$ N-back interaction tests.

Outcome	Global interaction p	Significant	Significant interaction terms
Accuracy	< .001	Yes	2
Reaction Time	= .067	No	2

Decision Rules

- H4 is supported if the Emotion $\times$ N-back interaction is statistically significant for both primary outcomes.
- H4 is partially supported if the interaction is statistically significant for only one primary outcome.
- H4 is not supported if neither global interaction test is statistically significant.
- Simple-effect analyses are used to characterize the load-dependent pattern and are not treated as substitutes for the global interaction tests.

Interpretation

Hypothesis 4 was partially supported.

The central prediction of Hypothesis 4 was that emotional effects on working-memory performance would depend on cognitive load rather than remaining constant across the 1-back through 4-back levels.

The interaction analyses indicate whether this load-dependent modulation occurred independently for accuracy and reaction time. Significant global interactions provide evidence that the magnitude or direction of the emotional-condition effect changed as task demands increased.

The simple-effect analyses clarify this interaction by identifying the specific n-back levels and emotional contrasts responsible for the overall pattern. Accordingly, the H4 results should be interpreted in terms of both the global interaction tests and the level-specific contrasts, rather than from isolated interaction coefficients alone.

Hypothesis Assessment

Conclusion: Hypothesis 4 was partially supported. The results indicate whether the influence of emotional condition on working-memory performance changed as cognitive load increased from 1-back to 4-back.